

References

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Poster No 70

Learning to control the visual cortex and enhancing visual attention via fMRI-based Neurofeedback

Riccardo Galli¹, Lucas Peek¹, Soraya Brosset¹, Frédéric Grouiller¹, Patrik Vuilleumier¹

¹University of Geneva, Geneva, Switzerland

Introduction: Attention is a crucial cognitive function allowing us to select pertinent sensory information while ignoring irrelevant stimuli in the environment¹. This capacity emerges from top-down mechanisms involving bilateral fronto-parietal networks that interact with early visual areas⁷. Following frontal or parietal brain lesions and therefore disruption in this system, peculiar conditions may emerge, such as unilateral spatial neglect (USN): a syndrome denoted by impaired awareness of stimuli presented in the visual field contralateral to the lesion site, in absence of pure sensorial or motor losses. One functional explanation of this condition might rely on abnormal biases in top-down regulation of sensory pathways from higher-level attentional networks towards early visual areas⁸. Neuromodulation and up-regulation of such preserved sensory areas have proved to account for partial restoration of this balance^{3,4,6} and improvement in clinical symptoms⁵. In particular, functional MRI (fMRI) based real-time neurofeedback (NFB)^{3,4,5,6} represents a promising and effective neuromodulation tool. However, specific mechanisms underlying successful modulation of visual areas via fMRI NFB are still unclear. For this reason, we couple the spatial precision of fMRI with the temporal resolution of EEG in EEG-MRI multimodal imaging fashion during NFB training to unravel structural and functional correlates of such learning process in the brain. Results from this study will further help develop an informed EEG-NFB based protocol to apply in clinical context for USN rehabilitation.

Methods: Following a double-blind randomized clinical trial routine, we train 30 participants to upregulate either left (N = 15) or right visual cortex (VC) over the course of 2 NFB training sessions, while a control group (N = 30) receives a sham feedback (i.e. a feedback sampled from another participant's brain activity). We then investigate the effects of NFB on behavior using several computerized tasks for visual attention as well as multiple neural (MRI, EEG) measures. Furthermore, we address the question of how successful NFB modulation of the visual cortex takes place in the brain at high spatial and temporal resolution thanks to multimodal EEG-MRI imaging applied during the second session of NFB. Finally, baseline EEG-MRI measures (resting-state fMRI, EEG, Diffusion Tensor Imaging) allow us to investigate possible predictive biomarkers of a successful NFB intervention at single subject level. NFB is performed using an open-source Python/Matlab based software (OpenNFT²) in combination with an in-house-developed Matlab based software for NFB sessions preparation (prepNFB⁹).

Results: Results show significant training effects in real-time estimated brain activity across NFB sessions in the experimental group training right VC (Fig.1, * denotes cluster at $p < .05$). Concomitant EEG time-frequency topographies during the second session of NFB show how such learning could be explained by beta and alpha band neuronal oscillations modulation. Furthermore, whole-brain analysis shows recruitment of higher-level brain areas generally involved in attentional processes (Fig.2, $p < .001$ unc), such as the Middle Frontal Gyrus (MFG), during neurofeedback regulation. Finally, such neural changes seem to differentially affect behavioral responses, biasing accuracy towards the contralateral visual field (compared to trained visual cortex) during a visual search task.

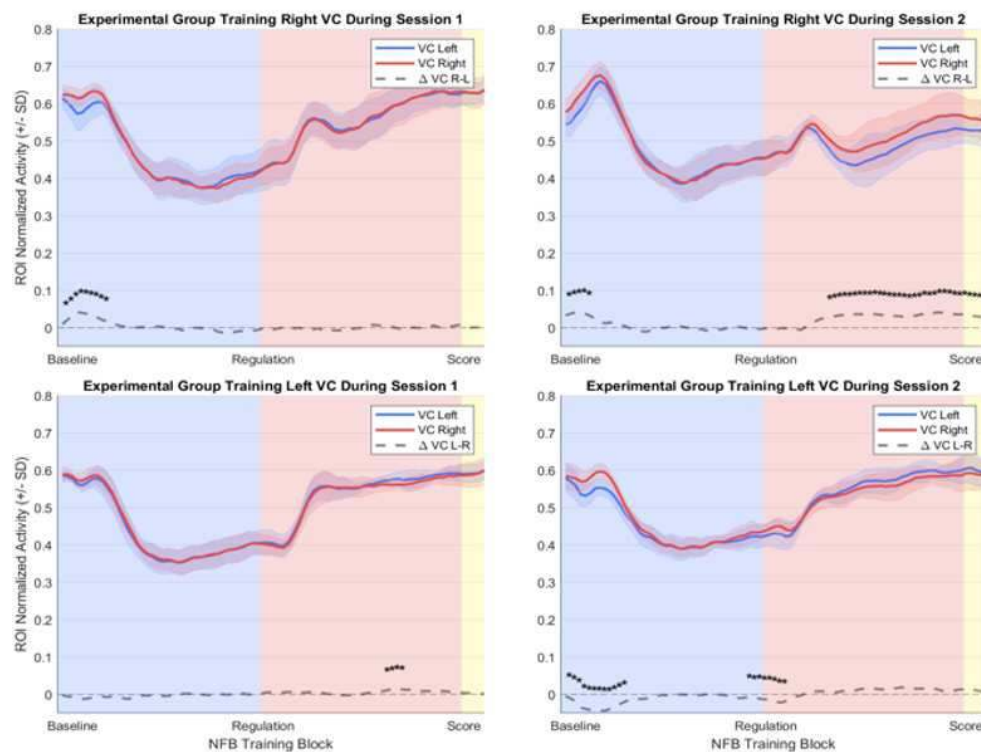
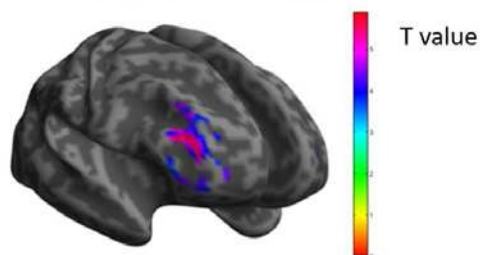


Figure 1

Exp Regulation – Baseline RVC



Exp Regulation – Baseline LVC

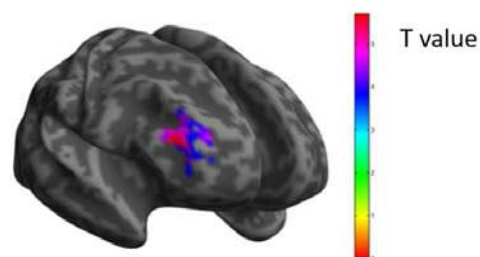


Figure 2

Conclusions: Participants appear able to learn how to regulate their occipital cortex activity thanks to real-time NFB and improve, according to training direction, behavioral performances in visual attention task(s). At the same time, such learning process might be reflected in modulation of alpha/beta brain frequencies and recruitment of higher-level attentional hubs.

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